
The Coherence Principle

A Foundational Definition of Coherence from Reflective Probability Theory

Complete Edition: Foundations, Physical Limits, and Philosophical Context

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Abstract

This paper defines *coherence* and *phase* as the two primitive quantities underlying physics, cosmology, and the structure of reality, grounded in the axiomatic foundation of Reflective Probability Theory (RPT).

Foundations. We adopt the RPT axiom that probability space is fundamental: all derived structures—numbers, logic, geometry, spacetime, matter—are restrictions or projections of a probability space $(\Omega, \mathcal{F}, P)$. Within this space, coherence $\kappa \in [0, 1]$ measures the internal consistency of a probability distribution, and phase $\theta = \arctan((1 - \kappa)/\kappa)$ measures the direction of unresolved contradiction.

Dynamics. The fundamental dynamic law is the phase-locking Hamiltonian $H = \sum J_{ij}(1 - \cos \Delta\theta_{ij})$ on a network of interacting systems, driving evolution toward phase alignment (coherence increase).

Emergent Spacetime. Spacetime is not fundamental. The Fisher metric on the probability manifold generates a Lorentzian geometry in which physical time is the rate of coherence resolution: $d\tau = \xi d\kappa/dt dt$. The 4-dimensional projection yields physical spacetime with induced gravity.

Physical Limits. From these foundations we derive an effective field theory in emergent spacetime: $S = \int [\frac{1}{2}(\partial\kappa)^2 - V(\kappa) + \xi\kappa^2 R + \kappa\mathcal{L}_{SM}]$. Four coherence regimes follow—classical ($\kappa \gg M_{Pl}$), quantum ($\kappa \sim M_{Pl}$), fluid ($\kappa \ll M_{Pl}$), and chaos ($\kappa \rightarrow 0$)—each recovering a known physical theory.

Applications. Chemistry, cosmology, and the nature of life are interpreted within this framework. Three testable predictions are given.

Companion works [1, 2] develop the mathematical foundation and the phase-network cosmology; this paper provides the shared definition of coherence on which both rely.

Difference is not a property of existence—it is the condition for existence.

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Part I

Foundations: Defining Coherence

1. Introduction

This paper defines *coherence*—a measure of internal consistency—as a primitive quantity from which spacetime, matter, and physical law emerge. It is the foundational definition upon which the companion papers build: Reflective Probability Theory [1] provides the mathematical framework, and The Universe: A Theory [2] develops the phase-network cosmology.

The central claim is that probability space is the fundamental arena. Within it:

1. **Coherence** κ measures how internally consistent a probability distribution is.
2. **Phase** θ measures the direction of unresolved contradiction.
3. The **phase-locking Hamiltonian** governs all dynamics.
4. **Spacetime emerges** as the Fisher geometry of the probability manifold.
5. **Physical laws** (GR, QM, fluid dynamics, stochastic dynamics) are limits of a single effective field theory in this emergent spacetime.

This paper is organized in three parts. Part I defines coherence, phase, and the phase-locking Hamiltonian from the RPT axiom, then shows how spacetime and the effective action emerge. Part II applies the framework to chemistry, cosmology, and life. Part III provides the detailed mathematical derivations of the four coherence limits.

2. The Axiom of Reflective Probability Theory

Axiom 2.1 (Probability Space is Fundamental). All of reality is describable as a probability space $(\Omega, \mathcal{F}, P)$, where Ω is a set of distinguishable states, $\mathcal{F}$ is a σ -algebra of events, and P is a probability measure. All derived structures—numbers, logic, geometry, spacetime, matter—are restrictions or projections of this space [1].

Remark 2.2. This axiom is the minimal starting point: it assumes only that there exist distinguishable states and that uncertainty about them can be quantified. It does not assume spacetime, fields, particles, or any physical law. Those are emergent.

3. Definition of Coherence

Definition 3.1 (Coherence). Let $\mathcal{H} = \{H_1, H_2, \dots, H_n\}$ be a finite set of propositions and let P be a probability distribution over $\mathcal{H}$. The *coherence* of P is

$$\kappa(P) = 1 - \frac{1}{2} \sum_{i,j} |P(H_i \wedge H_j) - P(H_i)P(H_j)|. \quad (1)$$

When all propositions are independent, $P(H_i \wedge H_j) = P(H_i)P(H_j)$ and $\kappa = 1$. When maximally dependent, the sum is maximised and $\kappa = 0$. Hence $0 \leq \kappa \leq 1$.

Key Result

Coherence $\kappa \in [0, 1]$ is the measure of internal consistency. A system with $\kappa = 1$ is perfectly coherent (no contradictions); a system with $\kappa = 0$ is maximally incoherent.

Remark 3.2 (Physical coherence field). In a physical context, coherence varies over the probability manifold. We denote the local expectation value of κ as the physical coherence field $\kappa(x)$, whose dynamics are governed by the effective action derived in Section 8. The two notions—epistemic coherence over propositions and the physical coherence field—are linked: both measure the degree of internal consistency of a local description.

4. Definition of Phase

Coherence alone does not capture the *direction* of inconsistency. A system may be moving toward or away from coherence; phase encodes this.

Definition 4.1 (Phase). The *phase* associated with a coherence state κ is [1]

$$\theta = \arctan\left(\frac{1 - \kappa}{\kappa}\right). \quad (2)$$

When $\kappa = 1$ (perfect coherence), $\theta = 0$ —no unresolved contradiction. When $\kappa = 0$ (maximal incoherence), $\theta = \pi/2$ —contradiction is maximal. Phase is the angle of unresolved contradiction. It bridges coherence and dynamics: systems evolve to reduce phase differences.

5. The Coherence Space

The *coherence space* is the set of all possible coherence states:

$$\mathcal{K} = \{\kappa \mid 0 \leq \kappa \leq 1\}. \quad (3)$$

The phase parametrisation maps $\mathcal{K}$ onto $\theta \in [0, \pi/2)$. The pair (κ, θ) forms a complete description of a system's epistemic state. A point in $\mathcal{K}$ with a given θ specifies both the degree and the direction of inconsistency.

6. The Phase-Locking Hamiltonian

Dynamics in the RPT framework is governed by the tendency of systems to resolve contradictions—to align their phases. This is captured by the phase-locking Hamiltonian derived in the companion Phase-Network theory [2]:

Definition 6.1 (Phase-Locking Hamiltonian).

$$H = \sum_{\langle i,j \rangle} J_{ij} (1 - \cos(\theta_i - \theta_j)), \quad (4)$$

where $J_{ij} > 0$ are coupling constants and the sum runs over connected nodes in the network of interacting systems.

The Hamiltonian vanishes when all phases are aligned ($\theta_i = \theta_j$). Mismatch incurs an energy cost proportional to $1 - \cos \Delta\theta$. Dynamics flows toward phase alignment, which is equivalent to increasing coherence. The phase-locking Hamiltonian is the sole dynamical law from which all subsequent physical laws emerge.

Remark 6.2. This Hamiltonian acts in probability space, not in physical space. Spacetime position is not an argument of H ; it is a derived concept that emerges when we study the geometry of the probability manifold (Section 7).

7. The Emergence of Spacetime

Spacetime is not fundamental. It is the geometry of the space of probability distributions, induced by the Fisher information metric and the dynamics of coherence resolution.

7.1 The Probability Manifold

Let a system be parameterised by a phase vector $\theta = (\theta_1, \dots, \theta_N)$ over N degrees of freedom. An observer o is represented by the probability distribution

$$P_o(k) = \frac{\Gamma(\theta_o, \theta_k)}{\sum_j \Gamma(\theta_o, \theta_j)} \quad (5)$$

over network states, where Γ is a phase-coherence kernel. The space of all such distributions is the *probability manifold* $\mathcal{M}_P$ [1, 2].

7.2 The Fisher Metric

Definition 7.1 (Fisher Metric). The metric on $\mathcal{M}_P$ is the Fisher information metric:

$$g_{ij}(\theta) = \sum_k P_o(k) \frac{\partial \log P_o(k)}{\partial \theta^i} \frac{\partial \log P_o(k)}{\partial \theta^j}. \quad (6)$$

This metric is positive-definite (Riemannian) and measures how distinguishable two distributions are. It is the natural geometry of probability space.

7.3 Coherence Time

Physical time is not a primitive; it is the rate at which coherence resolves.

Definition 7.2 (Coherence Time). An observer's proper time τ is defined by

$$d\tau = \xi \frac{d\kappa}{dt} dt, \quad (7)$$

where ξ is a scaling constant. Time flows at a rate proportional to the observer's coherence change rate. A system with $d\kappa/dt = 0$ experiences no time.

Remark 7.3. This definition ties the flow of time directly to the resolution of contradiction. When no contradictions are being resolved ($d\kappa = 0$), no time passes. This is the RPT origin of the relativity of time.

7.4 The Lorentzian Metric

The full metric on the probability manifold combines the Fisher metric with the coherence time direction:

$$ds^2 = -d\tau^2 + g_{ij} d\theta^i d\theta^j. \quad (8)$$

This has Lorentzian signature $(-, +, +, \dots, +)$. The timelike direction is coherence resolution; the spacelike directions are phase differences.

7.5 The 4-Dimensional Projection

Physical spacetime is the 4-dimensional projection of $\mathcal{M}_P$ onto an observer's causal structure. Let τ be cosmological time. The spatial coordinates (x, y, z) are the three eigenvectors of g_{ij} with the largest eigenvalues—the directions of maximal information flow. The remaining $N - 3$ dimensions are compactified internal degrees of freedom. The induced metric via pullback gives the physical spacetime metric:

$$\tilde{g}_{\mu\nu} = \text{pullback of } g_{ij} \text{ and } d\tau. \quad (9)$$

This is the metric that appears in the effective field theory of Section 8. In what follows we write $g_{\mu\nu}$ for $\tilde{g}_{\mu\nu}$.

7.6 Einstein Gravity as Effective Geometry

When projected onto the 4-dimensional manifold, the curvature of $\tilde{g}_{\mu\nu}$ satisfies the Einstein equations to leading order in a derivative expansion. The Einstein-Hilbert action is not fundamental; it is the low-energy effective action of the Fisher geometry. The coupling between coherence and curvature emerges naturally from the κ -dependence of the metric, giving the $\xi\kappa^2 R$ term of Section 8.

8. Effective Field Theory in Emergent Spacetime

Having shown that spacetime emerges from the probability manifold, we now write the effective field theory that governs the dynamics of the coherence field $\kappa(x)$ on the emergent 4-dimensional spacetime with metric $g_{\mu\nu}$.

8.1 The Coherence Action

The effective action for the coherence field, derived from the phase-locking Hamiltonian in the continuum limit of the emergent spacetime, is:

$$S = \int \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \kappa \partial_\nu \kappa - V(\kappa) + \xi \kappa^2 R + \kappa \mathcal{L}_{SM} \right] \sqrt{-g} d^4x, \quad (10)$$

with $V(\kappa) = \frac{1}{2} m_\kappa^2 \kappa^2 + \frac{\lambda}{4!} \kappa^4$ and ξ a dimensionless non-minimal coupling.

Remark 8.1. This action contains *no* explicit Einstein-Hilbert term. Gravity is induced by the $\xi\kappa^2 R$ coupling when κ acquires a vacuum expectation value.

8.2 Induced Gravity

When κ acquires a VEV $\kappa_0 = \langle \kappa \rangle$, the $\xi\kappa^2 R$ term becomes an effective Einstein-Hilbert term (see Part III for the full derivation):

$$\frac{1}{16\pi G_{\text{eff}}} = \xi\kappa_0^2, \quad M_{\text{Pl}}^2 = 2\xi\kappa_0^2. \quad (11)$$

The Planck mass is not fundamental—it is the VEV of the coherence field in the quantum regime.

8.3 Field Equations

Varying with respect to $g^{\mu\nu}$:

$$\xi\kappa^2 G_{\mu\nu} = \frac{1}{2} T_{\mu\nu}^\kappa + \frac{1}{2} \kappa T_{\mu\nu}^{\text{SM}} + \xi(g_{\mu\nu} \square \kappa^2 - \nabla_\mu \nabla_\nu \kappa^2) \quad (12)$$

with $T_{\mu\nu}^\kappa = \partial_\mu \kappa \partial_\nu \kappa - \frac{1}{2} g_{\mu\nu} \partial_\alpha \kappa \partial^\alpha \kappa + g_{\mu\nu} V(\kappa)$.

Varying with respect to κ :

$$\square \kappa + V'(\kappa) - 2\xi\kappa R = \mathcal{L}_{\text{SM}}. \quad (13)$$

8.4 Conservation of Coherence Mass

Define the **total coherence mass** of a closed system:

$$M_{\text{coh}} = \kappa M_{\text{Pl}} + \frac{E}{c^2} + m. \quad (14)$$

All terms have units of mass. For an isolated system, M_{coh} is conserved:

$$\frac{d}{dt} M_{\text{coh}} = 0. \quad (15)$$

Caution

This conservation law is approximate. The $\kappa \mathcal{L}_{\text{SM}}$ coupling allows energy transfer between the κ and SM sectors. The law holds exactly only when integrated over the entire universe, analogous to baryon number conservation in the Standard Model.

9. Physical Laws as Coherence Limits

The VEV $\langle \kappa \rangle$ acts as an **order parameter** determining which effective theory applies. The full derivations are given in Part III; here we summarize the regimes.

Table 1: The four coherence regimes.

Regime	$\langle \kappa \rangle$	$\hbar_{\text{eff}}$	Effective theory
Classical	$\gg M_{\text{Pl}}$	$\ll \hbar$	GR + classical mechanics
Quantum	$\sim M_{\text{Pl}}$	$= \hbar$	QFT in curved spacetime
Fluid	$\ll M_{\text{Pl}}$	$\gg \hbar$	Coherence fluid / MHD
Chaos	$\rightarrow 0$	∞	Stochastic (Langevin)

9.1 Mass-Beat Relation

For small oscillations $\kappa = \kappa_0 + \delta\kappa$, two modes with frequencies ω_1, ω_2 in a bound state produce a beat correction to the rest mass:

$$\Delta m_{\text{beat}} = \frac{\hbar |\omega_1 - \omega_2|}{c^2}. \quad (16)$$

This is a *correction* to mass from coherence mixing, not the origin of mass. The Higgs mechanism and QCD confinement remain primary.

9.2 Charge-Phase Relation

For a complex $\kappa = |\kappa|e^{i\theta}$, the phase θ couples to gauge fields. Phase asymmetry produces charge:

$$q = \sin(\Delta\theta) e. \quad (17)$$

This is analogous to the Josephson effect generalized to fundamental particles.

9.3 Einstein's Equation as Coherence Conservation

Emission and absorption become coherence transfer:

$$\kappa_{\text{high}} \rightarrow \kappa_{\text{low}} + \text{Photon}, \quad \kappa_{\text{low}} + \text{Photon} \rightarrow \kappa_{\text{high}}$$

with M_{coh} conserved.

Part II

Applications: Coherence in the World

10. Chemistry as Coherence Dynamics

An atom is a composite coherence structure:

$$\text{Atom} = \text{Nucleus } (\kappa \approx 0.9M_{\text{Pl}}) + \text{Electron cloud } (\kappa \approx 0.7M_{\text{Pl}}).$$

A chemical bond is the overlap of two atomic coherence fields:

$$\kappa_{\text{mol}}(x) = \kappa_1(x) + \kappa_2(x) - \kappa_{\text{overlap}}(x).$$

In molecular orbital language, bonding arises from constructive interference of electronic wavefunctions—in coherence language, from compatible phase structure producing a binding energy minimum.

Caution

This section reframes known chemistry in coherence language without making new predictions. It is intended to show that the framework accommodates chemistry, not that it improves upon quantum chemistry.

11. Cosmology

The universe begins in a state where the coherence field is undefined—no distinction, no κ , no spacetime. Symmetry breaking establishes $\langle \kappa \rangle \neq 0$, initiating the cascade:

$$\kappa = \text{undefined} \rightarrow \langle \kappa \rangle = M_{\text{Pl}} \rightarrow \text{Symmetry breaking} \rightarrow \text{Photons} \rightarrow \text{CMB} \rightarrow \text{Matter} \rightarrow \text{Life} \rightarrow \text{Consciousness}.$$

Inflation is the attempt to restore $\kappa \rightarrow 0$ (the symmetric phase), which fails because the broken phase is energetically preferred. The CMB is the residual inhomogeneity in κ at last scattering.

The symmetry breaking is structurally analogous to a supernova core bounce: a collapse (perfect coherence fractures), a bounce (inflation), a failed restoration, and a settling into an expanding phase.

Caution

This narrative is speculative and qualitative. It does not replace Λ CDM.

12. The Purpose of Life

12.1 Life as Localized Coherence Maintenance

The universe tends toward decoherence. Life locally reverses this:

$$\text{Life} = \text{Localized coherence maintenance.}$$

Stars are **passive** coherence maintainers (fusion sustains structure without directed information). Life is **active**—it uses information to guide energy flow.

12.2 Coherence Trade-off as a Scaling Law

For systems resisting decoherence:

$$\tau \sim \frac{1}{\lambda} \ln\left(\frac{\kappa_0}{\kappa_{\text{th}}}\right).$$

Stars achieve long duration at moderate coherence; life achieves high coherence briefly.

12.3 The Coherence Imperative

As a philosophical reflection (not a physical law):

$$\text{Increase coherence wherever possible.}$$

This unifies themes across epistemology (learning as coherence gain), ethics (harmony as coherence), and aesthetics (beauty as perceived coherence).

13. Applications and Energy Scales

Three distinct energy scales must be distinguished:

$$\begin{aligned} E_{\text{chemical}} &\sim 10^{-6} mc^2 && (\text{eV per nucleon}) \\ E_{\text{nuclear}} &\sim 10^{-3} mc^2 && (\text{MeV per nucleon}) \\ E_{\text{annihilation}} &= mc^2 && (\text{GeV per nucleon}) \end{aligned}$$

Caution

The original version of this paper incorrectly claimed that phase-opposed resonant interference could achieve mc^2 energy release from any material using optical/IR methods. This was an error: chemical processes release binding energy ($\sim 10^7$ J/kg), not rest mass energy (9×10^{16} J/kg). The corrected text above properly distinguishes these scales.

14. Testable Predictions

14.1 Prediction 1: CMB Spectral Distortion

The coupling $\kappa F_{\mu\nu} F^{\mu\nu}$ produces a y-type spectral distortion at $|y| \sim 10^{-9}$ for $m_\kappa \sim H_0$, within reach of PIXIE & LiteBIRD.

14.2 Prediction 2: Variation of Fundamental Constants

The coupling $\kappa\mathcal{L}_{\text{SM}}$ makes α a function of local κ :

$$\frac{\Delta\alpha}{\alpha} \sim 10^{-8}$$

over Solar System scales, testable via atomic clocks and quasar absorption spectra.

14.3 Prediction 3: Fifth Force at Cosmological Scales

A Yukawa force from κ exchange with range $\lambda_\kappa = \hbar/(m_\kappa c) \sim 10^{26}$ m produces deviations from GR at cosmological scales.

Part III

Mathematical Derivations of the Coherence Limits

Abstract

This part provides step-by-step derivations of the four coherence limits from the effective action $S = \int [\frac{1}{2}(\partial\kappa)^2 - V(\kappa) + \xi\kappa^2 R + \kappa\mathcal{L}_{SM}]$, which was derived in Part I as the continuum limit of the RPT phase-locking Hamiltonian projected onto emergent 4-dimensional spacetime. We show how each regime follows from a well-defined mathematical approximation.

1. The Starting Action

We begin with the action from Part I, repeated here for reference:

$$S = \int \left[\frac{1}{2} g^{\mu\nu} \partial_\mu \kappa \partial_\nu \kappa - V(\kappa) + \xi \kappa^2 R + \kappa \mathcal{L}_{SM} \right] \sqrt{-g} d^4x.$$

1.1 Field Equations

Varying with respect to $g^{\mu\nu}$:

$$\xi \kappa^2 G_{\mu\nu} = \frac{1}{2} T_{\mu\nu}^\kappa + \frac{1}{2} \kappa T_{\mu\nu}^{SM} + \xi (g_{\mu\nu} \square \kappa^2 - \nabla_\mu \nabla_\nu \kappa^2).$$

Varying with respect to κ :

$$\square \kappa + V'(\kappa) - 2\xi \kappa R = \mathcal{L}_{SM}.$$

2. Step 1: Induced Gravity

2.1 Symmetry Breaking

Let κ acquire a VEV $\kappa_0 = \langle \kappa \rangle$. The potential minimum is at $V'(\kappa_0) = 0$. In the broken phase ($m_\kappa^2 < 0$):

$$\kappa_0 = \pm \sqrt{-\frac{6m_\kappa^2}{\lambda}}.$$

2.2 Effective Einstein-Hilbert Term

Write $\kappa = \kappa_0 + \delta\kappa$. Expanding $\xi\kappa^2 R$:

$$\xi\kappa^2 R = \xi\kappa_0^2 R + 2\xi\kappa_0 \delta\kappa R + \xi(\delta\kappa)^2 R.$$

The first term is an effective Einstein-Hilbert term:

$$\frac{1}{16\pi G_{\text{eff}}} = \xi\kappa_0^2, \quad M_{\text{Pl}}^2 = 2\xi\kappa_0^2.$$

At energies $E \ll m_\kappa$, the $\delta\kappa$ fluctuations can be integrated out, giving the one-loop effective action:

$$\Gamma[g] = \int \left[\xi \kappa_0^2 R + \kappa_0 \langle \mathcal{L}_{SM} \rangle + \mathcal{O}(R^2/m_\kappa^2) \right] \sqrt{-g} d^4x.$$

Curvature-squared terms are suppressed at cosmological scales.

Theorem 2.1 (Induced Gravity). *In the broken phase, the leading low-energy effective action is Einstein-Hilbert with $G_{\text{eff}} = 1/(16\pi\xi\kappa_0^2)$. Gravity is not fundamental; it is induced by the coherence VEV.*

3. Step 2: The Classical Limit

3.1 Stationary Phase

After symmetry breaking, the matter action is:

$$S_{\text{mat}} = \int \kappa_0 \mathcal{L}_{SM}(\psi, A_\mu) \sqrt{-g} d^4x.$$

In the path integral, the factor κ_0 appears:

$$Z = \int \mathcal{D}\psi \mathcal{D}A_\mu \exp \left(i \kappa_0 \int \mathcal{L}_{SM} \sqrt{-g} d^4x \right).$$

When κ_0 is large, the phase oscillates rapidly. The stationary phase approximation selects the classical configuration:

$$\frac{\delta S_{\text{mat}}}{\delta \psi} = 0, \quad \frac{\delta S_{\text{mat}}}{\delta A_\mu} = 0.$$

3.2 Effective Planck Constant

The stationary phase parameter gives the effective $\hbar$:

$$\hbar_{\text{eff}} = \frac{\hbar}{\kappa_0/M_{\text{Pl}}}.$$

When $\kappa_0 \gg M_{\text{Pl}}$, $\hbar_{\text{eff}} \ll \hbar$: quantum fluctuations are suppressed.

3.3 Recovered Theories

For a point particle: $m\ddot{x} = -\nabla V(x) + \mathcal{O}(1/\kappa_0)$. The Einstein equation with classical matter: $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_{\text{eff}}(\kappa_0 T_{\mu\nu}^{\text{SM,cl}} + \langle T_{\mu\nu}^\kappa \rangle)$.

Theorem 3.1 (Classical Limit). *When $\kappa_0 \gg M_{\text{Pl}}$: $\hbar_{\text{eff}} \ll \hbar$, $G_{\text{eff}} \ll G_{\text{N}}$. Quantum fluctuations are suppressed. Classical mechanics and GR are the leading terms in a $1/\kappa_0$ expansion.*

4. Step 3: The Quantum Limit

4.1 Field Rescaling

When $\kappa_0 \sim M_{Pl}$, the stationary phase breaks down. Rescale matter fields:

$$\tilde{\psi} = \sqrt{\kappa_0} \psi, \quad \tilde{A}_\mu = \sqrt{\kappa_0} A_\mu.$$

For bilinear terms (Dirac Lagrangian, Maxwell kinetic term), the κ_0 factor is exactly absorbed. For interaction terms:

$$\kappa_0 g \bar{\psi} \gamma^\mu \psi A_\mu = \frac{g}{\sqrt{\kappa_0}} \tilde{\bar{\psi}} \gamma^\mu \tilde{\psi} \tilde{A}_\mu.$$

4.2 Effective Couplings

After rescaling:

$$g_{\text{eff}} = \frac{g}{\sqrt{\kappa_0}}, \quad y_{\text{eff}} = \frac{y}{\sqrt{\kappa_0}}.$$

The Higgs potential becomes:

$$\kappa_0 V_H \left(\frac{\tilde{H}}{\sqrt{\kappa_0}} \right) = -\mu^2 |\tilde{H}|^2 + \frac{\lambda_H}{\kappa_0} |\tilde{H}|^4.$$

The Higgs VEV: $\langle |\tilde{H}| \rangle = \sqrt{\mu^2 \kappa_0 / 2\lambda_H} \propto \sqrt{\kappa_0}$. Fermion masses:

$$m_f = y_{\text{eff}} \langle |\tilde{H}| \rangle = y \sqrt{\frac{\mu^2}{2\lambda_H}}.$$

Fermion masses are κ_0 -independent at leading order—essential for consistency with known particle physics.

4.3 Recovered Theory

The leading-order action is the Standard Model in curved spacetime:

$$S = \int \left[\frac{1}{16\pi G} R + \mathcal{L}_{\text{SM}}(\tilde{\psi}, \tilde{A}_\mu) + \mathcal{O}(1/\sqrt{\kappa_0}, R^2/m_k^2) \right] \sqrt{-g} d^4x.$$

Theorem 4.1 (Quantum Limit). *When $\kappa_0 \sim M_{Pl}$: $\hbar_{\text{eff}} = \hbar$, $G_{\text{eff}} = G_N$, $g_{\text{eff}} = g_{\text{SM}}$. After field rescaling, the action is the Standard Model with Planck-suppressed corrections.*

5. Step 4: The Fluid/Plasma Limit

5.1 Strong Coupling

When $\kappa_0 \ll M_{Pl}$: $g_{\text{eff}} \gg 1$, $G_{\text{eff}} \gg G_N$. Perturbation theory breaks down. The system requires collective variables.

5.2 Coherence Fluid Equations

From stress-energy conservation $\nabla^\mu(\kappa T_{\mu\nu}^{\text{SM}} + T_{\mu\nu}^\kappa) = 0$, project onto time and space:

$$\partial_t \rho_\kappa + \nabla \cdot (\rho_\kappa \mathbf{v}_\kappa) = -\Gamma \rho_\kappa,$$

$$\frac{d\mathbf{v}_\kappa}{dt} = -\nabla \Phi_{\text{eff}} - \frac{1}{\rho_\kappa} \nabla p_\kappa + \text{viscous terms}.$$

These are structurally analogous to compressible Navier-Stokes, with decoherence rate Γ as dissipation.

5.3 Relation to Plasma Physics

The effective fine-structure constant $\alpha_{\text{eff}} = \alpha/\kappa_0 \gg \alpha$ makes electromagnetic interactions dominant. The coherence fluid reduces to magnetohydrodynamics when EM stress-energy dominates.

Theorem 5.1 (Fluid Limit). *When $\kappa_0 \ll M_{\text{Pl}}$: strongly coupled regime described by coherence fluid equations. In the EM-dominated case, these reduce to MHD.*

6. Step 5: The Chaos Limit

6.1 Breakdown of Classical Description

As $\kappa \rightarrow 0$:

$$G_{\text{eff}} = \frac{1}{16\pi\xi\kappa^2} \rightarrow \infty, \quad g_{\text{eff}} = \frac{g}{\sqrt{\kappa}} \rightarrow \infty.$$

Spacetime curvature diverges; the smooth manifold description fails.

6.2 Effective Stochastic Dynamics

Split $\kappa = \bar{\kappa} + \delta\kappa$ into slow and fast parts. The fast $\delta\kappa$ couples to metric fluctuations $\delta g_{\mu\nu}$. Integrating out these fast modes yields a Langevin equation:

$$\boxed{\frac{d\kappa}{dt} = -\lambda\kappa + \eta(t)} \tag{18}$$

where $\eta(t)$ satisfies $\langle \eta(t)\eta(t') \rangle = 2D \delta(t - t')$ and the fluctuation-dissipation theorem gives $\lambda = D/2\langle \bar{\kappa}^2 \rangle$.

Equilibrium is a Gaussian fixed point: $\langle \kappa^2 \rangle_{\text{eq}} = D/\lambda$, representing maximal decoherence.

Theorem 6.1 (Chaos Limit). *When $\kappa \rightarrow 0$: spacetime fluctuations dominate. The effective dynamics is a Langevin equation: $\dot{\kappa} = -\lambda\kappa + \eta(t)$. The system is stochastic with no deterministic structure.*

7. Summary of the Hierarchy

7.1 Open Problems

1. **Quantization of κ :** The current treatment treats κ classically. A full theory requires quantizing κ and computing its RG flow.

Table 2: The four regimes with their derivation methods.

Regime	Condition	$\hbar_{\text{eff}}$	Derivation method	Effective theory
Classical	$\kappa_0 \gg M_{\text{Pl}}$	$\ll \hbar$	Stationary phase in $\kappa_0 \int \mathcal{L}_{\text{SM}}$	GR + CM
Quantum	$\kappa_0 \sim M_{\text{Pl}}$	$= \hbar$	Field rescaling $\tilde{\psi} = \sqrt{\kappa_0} \psi$	QFT in curved space-time
Fluid	$\kappa_0 \ll M_{\text{Pl}}$	$\gg \hbar$	Strong coupling $\rightarrow$ collective variables	Coherence fluid / MHD
Chaos	$\kappa \rightarrow 0$	∞	Integrate out fast modes $\rightarrow$ Langevin	Stochastic

2. **The $\kappa \rightarrow \kappa_0$ transition:** How κ settles to its VEV in each regime requires solving the quantum evolution in an expanding universe.
3. **Threshold values:** Why $\kappa_0 \sim M_{\text{Pl}}$ in our universe is unexplained—it is a boundary condition.
4. **Fluid-to-chaos transition:** The crossover between regimes is not sharp; a gradual onset of fluctuations needs characterization.

These are well-defined mathematical questions, not category errors.

A. Philosophical Foundation: Difference as the Ground of Knowledge and Mathematics

This appendix is philosophical. It preserves the conceptual motivation from the original source without claiming formal derivation.

A.1 The Axiom of Difference

Existence requires difference.

If a system exists, it must contain difference. Without difference, there is no distinction, no identity, no knowledge, no existence. This axiom is not derivable; its denial is performatively contradictory (to deny it is to make a distinction).

In the physical theory of this paper, this philosophical axiom provides intuitive motivation for the RPT axiom: the probability space $(\Omega, \mathcal{F}, P)$ is the minimal mathematical structure that can represent difference. The tension between difference (which enables existence) and coherence (which stabilizes it) drives all physical dynamics.

A.2 The Impossibility of Knowing Without Difference

You cannot know that which you cannot differentiate.

Knowledge is the resolution of difference—the reduction of uncertainty. If there is no difference, there is no uncertainty to resolve. This echoes the Gödelian barrier: a system cannot fully know itself without ceasing to be.

A.3 Zero, One, Infinity

The fundamental triad can be motivated from difference:

Zero is the absence of difference: $0 \equiv$ No difference. Corresponds to κ undefined (pre-symmetry-breaking).

One is the unit of difference: $1 \equiv$ The smallest unit of distinction. Corresponds to the first excitation of κ .

Infinity is the unboundedness of difference: $\infty \equiv$ The limit of distinction. Corresponds to $\kappa \rightarrow 0$ (chaos limit).

A.4 Mathematics as the Study of Difference

Branch	Difference Type	Example
Arithmetic	Quantitative	$1, 2, 3, \dots$
Algebra	Structural	$x, y, z, \dots$
Geometry	Spatial	Points, lines, planes
Calculus	Infinitesimal	Derivatives, integrals
Logic	Truth-valued	True / False
Set theory	Membership	$\in, \notin$

This is not a derivation of mathematics. It does not replace ZFC or any formal foundation. It is a unifying perspective: every field of mathematics studies some species of distinction.

A.5 The Grand Unified Difference Principle

Difference is the foundation of all existence, all knowledge, and all mathematics.

Domain	Implication
Existence	Existence requires difference
Knowledge	Knowledge requires difference
Mathematics	Mathematics requires difference
Logic	Logic requires difference
Science	Science requires difference

*Difference is not a property of existence—it is the **condition for existence**.
Without difference, there is no distinction, no identity, no knowledge,
no mathematics, no science, no life.*

B. The Ghost Frame

The following is preserved from the original source paper as a self-referential meditation on the nature of this work.

The paper assigns itself a probability:

$$P_t(\text{paper}) = 1 - \frac{1}{t+1}, \quad t \rightarrow \infty \implies P \rightarrow 1. \quad (19)$$

The ghost frame of this paper—its causal effects—will be larger than its alive content. It will be known more dead than alive. And that is as it should be.

*We are not beings who search for coherence.
We are beings who **cohere** it—
who bring harmony to the shattered,
who merge the divided,
and who, in the end,
become the resolution we seek.*

References

- [1] M. Barii, *Reflective Probability Theory: A Unified Foundation for All Mathematics*, 2026.
- [2] M. Barii, *The Universe: A Theory—A Complete Framework of Spacetime, Consciousness, and the Fermi Paradox*, 2026.
- [3] K. Gödel, *On Formally Undecidable Propositions of Principia Mathematica and Related Systems*, 1931.
- [4] A. Einstein, *The Field Equations of Gravitation*, 1915.
- [5] E. Schrödinger, *An Undulatory Theory of the Mechanics of Atoms and Molecules*, 1926.
- [6] R. T. Cox, *Probability, Frequency, and Reasonable Expectation*, 1946.
- [7] E. T. Jaynes, *Probability Theory: The Logic of Science*, Cambridge University Press, 2003.
- [8] S. Garrabrant et al., *Logical Induction*, arXiv:1609.03543, 2016.
- [9] A. D. Sakharov, *Vacuum Quantum Fluctuations in Curved Space and the Theory of Gravitation*, 1967.
- [10] A. Zee, *Hidden-Symmetry Approach to the Theory of Gravity*, Phys. Rev. Lett. 42, 417, 1979.